Advancement in Data Clustering Algorithms

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Data Clustering

Definition : Clustering refers to unsupervised classification of a dataset into groups, such that elements within a group are similar to each other.

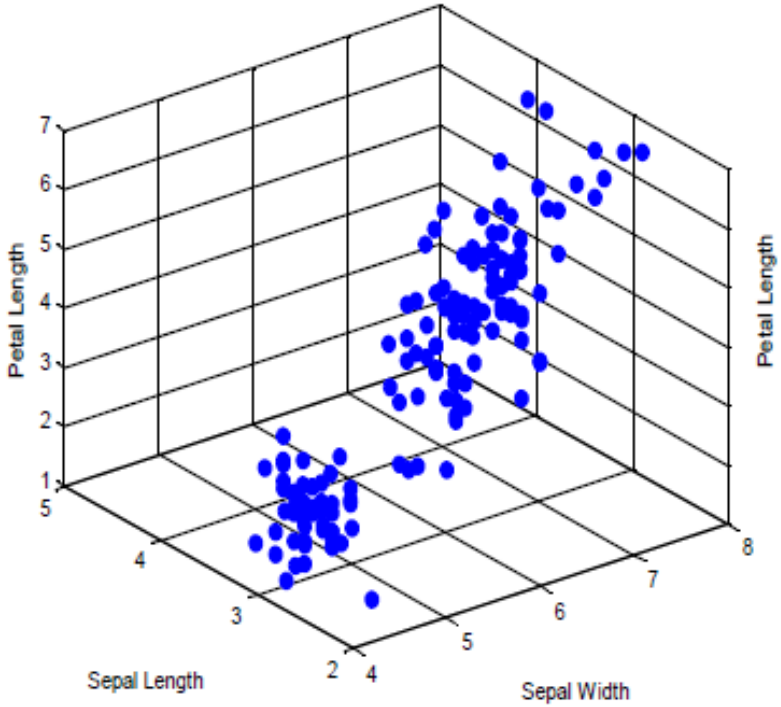
Example : IRIS Flower Dataset

150 samples (50 for each flower); 4 attributes (SL, SW, PL, PW)

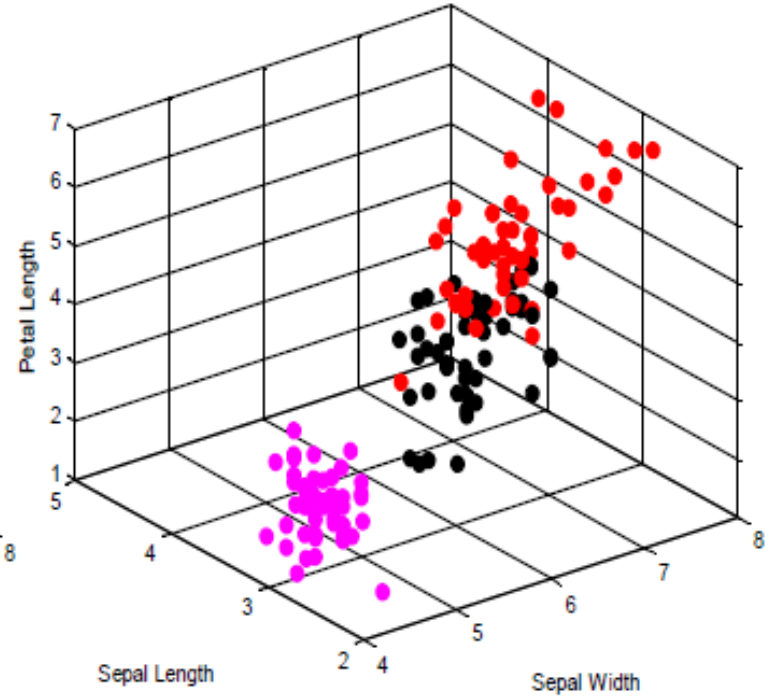
Setosa

Versicolor

Versinica



3D Plot of unlabeled IRIS dataset



3D Plot after cluster analysis

Partitional Clustering

A given dataset

$$\boldsymbol{P}_{N\times D} = \begin{bmatrix} p_{1,1} & p_{1,2} & \cdots & p_{1,D} \\ p_{2,1} & e_{2,2} & \cdots & p_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N,1} & p_{N,2} & \cdots & p_{N,D} \end{bmatrix}$$

Objective : Partition the dataset into M groups (clusters) such that

- Every cluster must contain at least one pattern

$$C_i \neq \phi \;\; \forall i \in 1, 2, \ldots M$$

- There should not be any common pattern between two clusters

$$C_i \cap C_j = \phi \;\; \forall i, j \in 1, 2, \ldots M \text{ and } i \neq j$$

- Dataset consists of sum of all the clusters

$$\bigcup_{i=1}^{M} C_i = P_{N\times D}$$

Clustering as an Optimization problem

Let Z_i represent center of cluster C_i. If F denotes a function then the clustering task is viewed as an optimization problem as

$$\underset{C_i}{\overset{Optimize}{}} \quad [F(P_{N\times D}, Z_i)] \quad \forall i = 1, 2, \ldots M$$

- Minimization of Euclidean Distance (K-Means Clustering)

$$\begin{aligned} \text{Minimize } F_1 &= \sum_{i=1}^{N} \sum_{j\in[1,M]} D(p_i, Z_j) \\ &= \sqrt{\sum_{i=1}^{N}(p_i - Z_1)^2} + \cdots + \sqrt{\sum_{i=1}^{N}(p_i - Z_M)^2} \end{aligned}$$

K-Means Clustering - Stuart Lloyd of Bell Labs in 1957

Initialized Cluster centres

Z1				Z2				Z3			
SL	SW	PL	PW	SL	SW	PL	PW	SL	SW	PL	PW
0.4	0.2	0.7	0.3								

IRIS Dataset

Sample	SL	SW	PL	PW					**Cluster Assignment**
P1	0.1	0.2	0.4	0.6	→	**D11**	**D12**	**D13**	C1
P2	0.3	0.1	0.5	0.8	→	**D12**	**D22**	**D23**	C3
:									
:									
:									
P150	0.8	0.9	0.2	0.4	→	**D150,1**	**D150,2**	**D150,3**	C2

Calculation of Updated Cluster centres

Cluster	SL	SW	PL	PW
C1	0.1	0.2	0.4	0.6
C1	0.3	0.1	0.5	0.8
:				
C1	0.6	0.8	0.4	0.5
Mean	∑SL(1:40) /40	∑SW(1:40) /40	∑PL(1:40) /40	∑PW(1:40) /40
Updated Z1	0.4	0.5	0.3	0.6

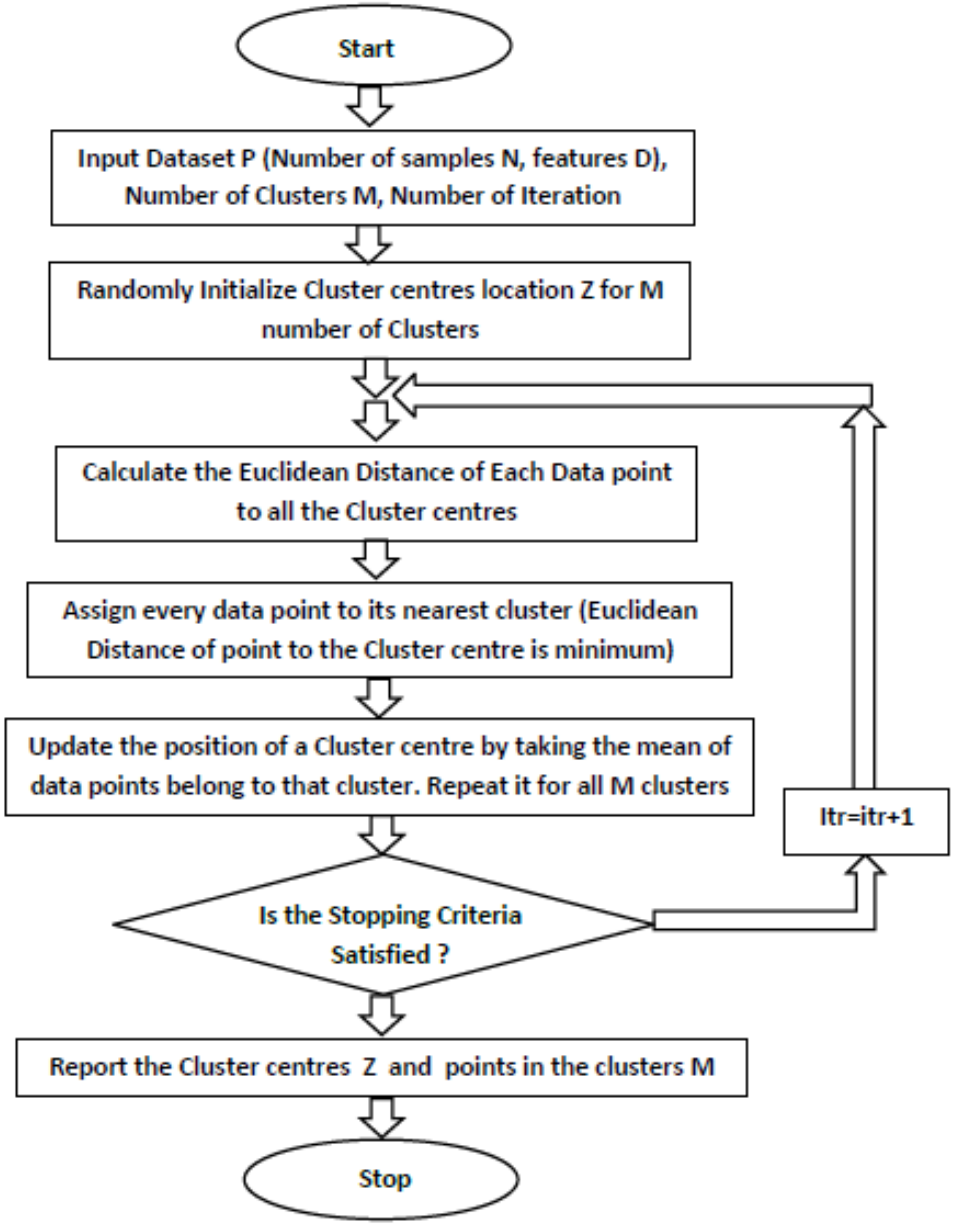
Total 40 points in Cluster C1

Similarly Update Cluster centres

Updated Z2			
SL	SW	PL	PW

Updated Z2			
SL	SW	PL	PW

K-Means Clustering FLOW-CHART



K-Medians clustering

- It is a variation of K-means clustering where instead of calculating the mean for each cluster to determine its centroid, one calculates median.

Example : Calculation of Median

- The median of a finite list of numbers can be found by arranging all the numbers from smallest to greatest.
- If there is an odd number of numbers, the middle one is picked. For example, consider the list of numbers
  1, 3, 3, 6, 7, 8, 9
  This list contains seven numbers. The median is the fourth of them, which is 6.
- If there is an even number of observations, then there is no single middle value; the median is then usually defined to be the mean of the two middle values. For example, in the data set
  1, 2, 3, 4, 5, 6, 8, 9
  the median is the mean of the middle two numbers: (4+5)/2=4.5

K-Medoids clustering or Partitioning Around Medoids

- K-medoids chooses data points as centers (medoids or exemplars) and can be used with arbitrary distances
- K-means the centre of a cluster is not necessarily one of the input data points (it is the average between the points in the cluster)

Definition : Medoids are representative objects of a data set or a cluster with a data set whose average dissimilarity to all the objects in the cluster is minimal.

Steps of Partitioning Around Medoids (PAM) [1] algorithm

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1 : A set of medoids is chosen at random.
2 : The distances to the other points are computed.
3 : Data are clustered according to the medoid they are most similar to.
4 : The medoid set is optimized via an iterative process.

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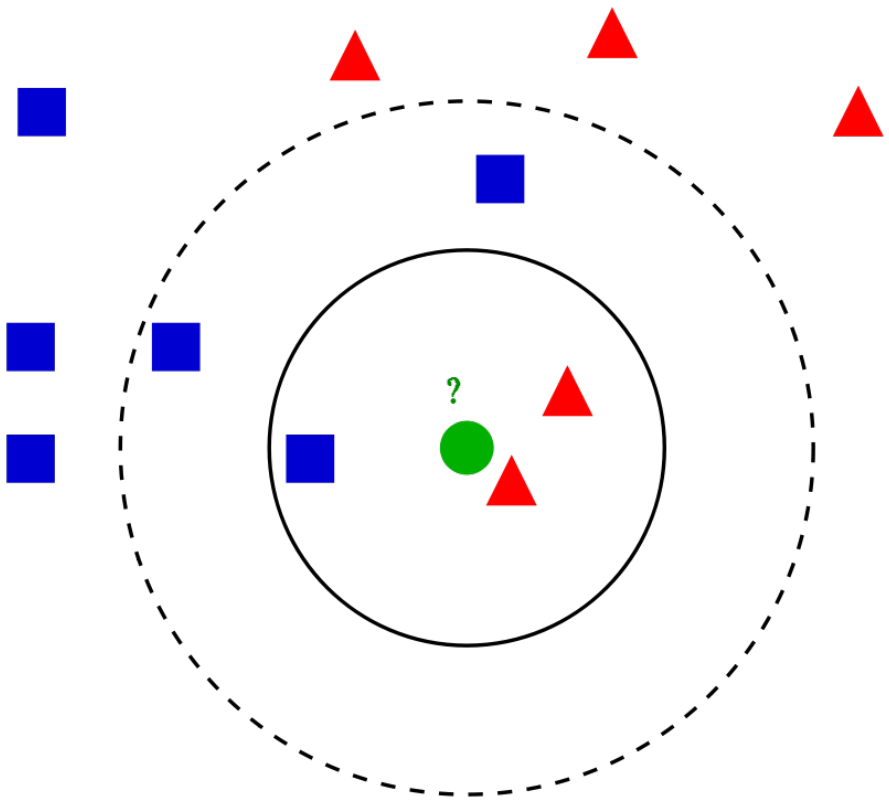
[1] Kaufman, L. and Rousseeuw, P.J. (1987), Clustering by means of Medoids, in Statistical Data Analysis Based on the L_1–Norm and Related Methods, edited by Y. Dodge, North-Holland, 405–416.

K-Nearest Neighbors (K-NN) algorithm

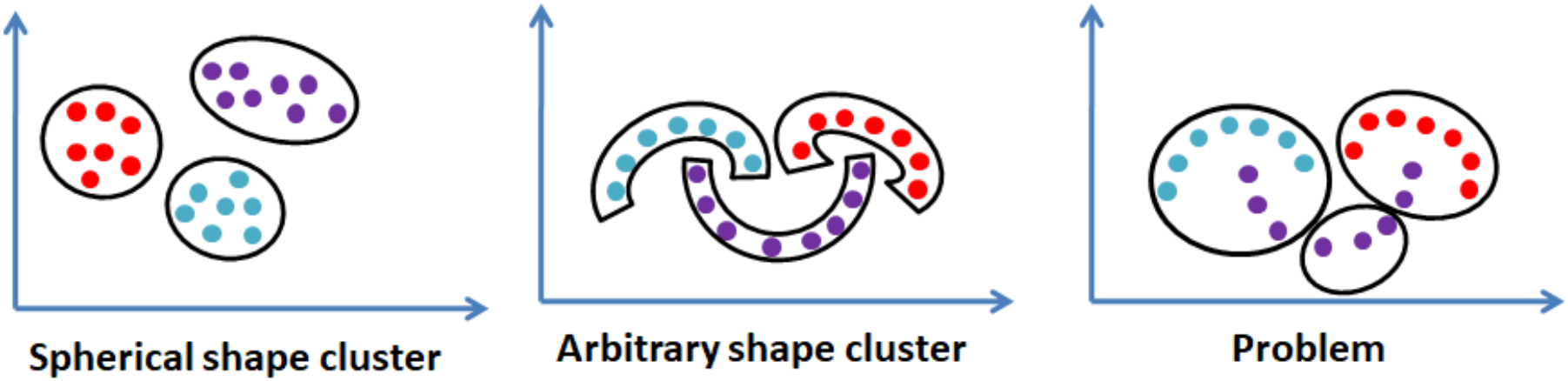


Example of K-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If K = 3 (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If K = 5 (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

Drawback of Partitional Clustering

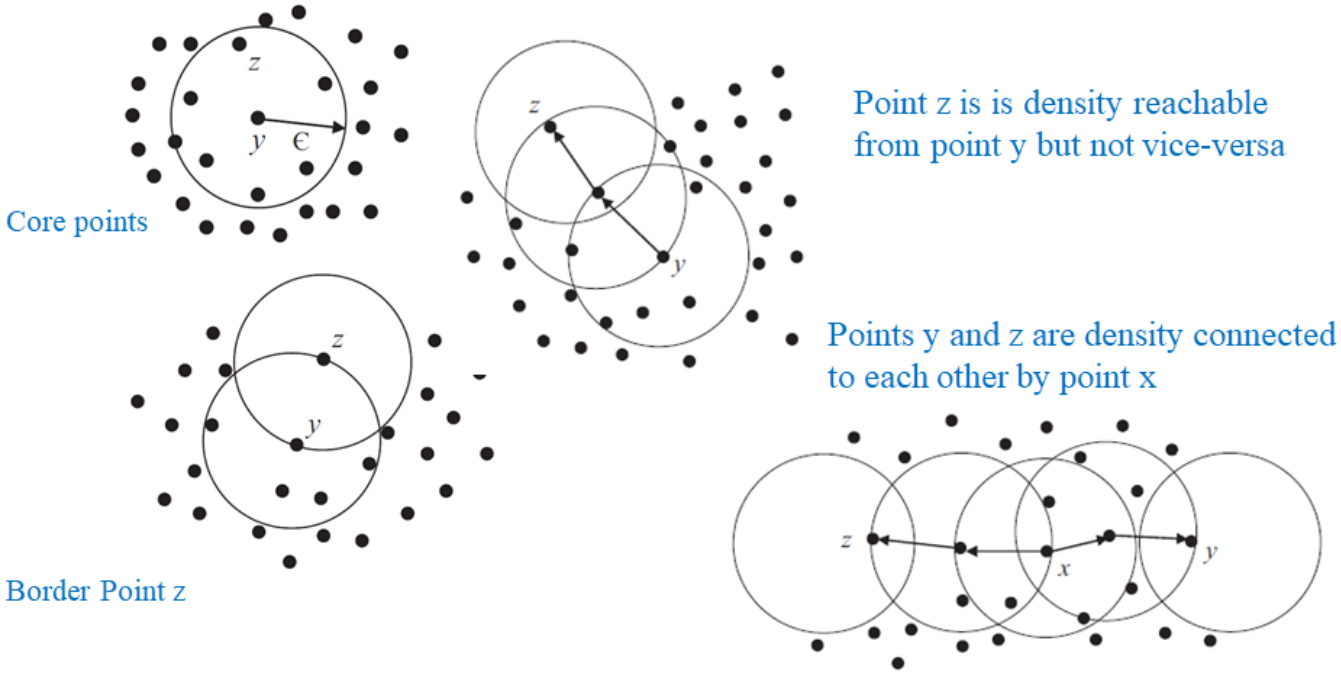
K-means algorithm

- Uses minimization of Eucliden distance as the similarity measure
- Effective for circular/spherical shape cluster



DBSCAN (Density Based Spatial Clustering of Applications with Noise)

- The algorithm is very popular and effective for arbitrary shape cluster.
- It involves two parameters : ϵ : radius of circle associated with a point and Mn-Pts : Minimum number of points in a circle.



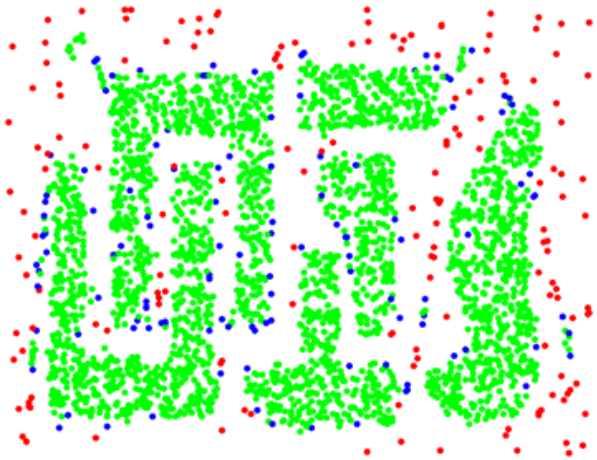
DBSCAN (Density Based Spatial Clustering of Applications with Noise)

1. Randomly select any point
2. Determine all the points which are density reachable from y as
   - If y is a core point, then y along with points in its ϵ neighborhood are defined as a cluster.
   - If y is a border point, then no points are density-reachable from y and the algorithm moves to the next unclassified point z.
3. Compute the distance of z from all of the established core points.
   - If distance of z from any core point is less than ϵ, then z can not be a core point. The algorithm discards such point and then moves to the next point.
   - Otherwise z is considered as a core point.
4. Steps 2-3 are continued all points are covered and algorithm end with small clusters.
5. Compute distance between clusters : d(C_1,C_2)=mind(y,z) : y ϵ C_1, z ϵ C_2
6. If d(C_1,C_2) $\leq \epsilon$ then merge the clusters C_1 and C_2.Repeat it for all small clusters.

DBSCAN (Density Based Spatial Clustering of Applications with Noise)

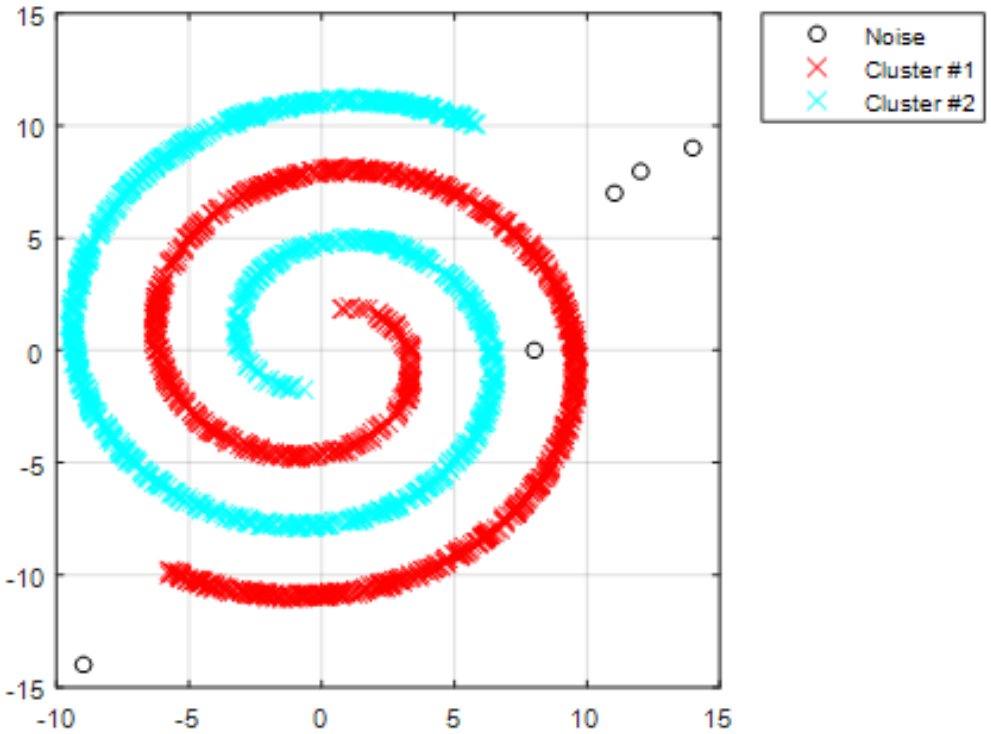
Original Points

Point types: core, border and noise

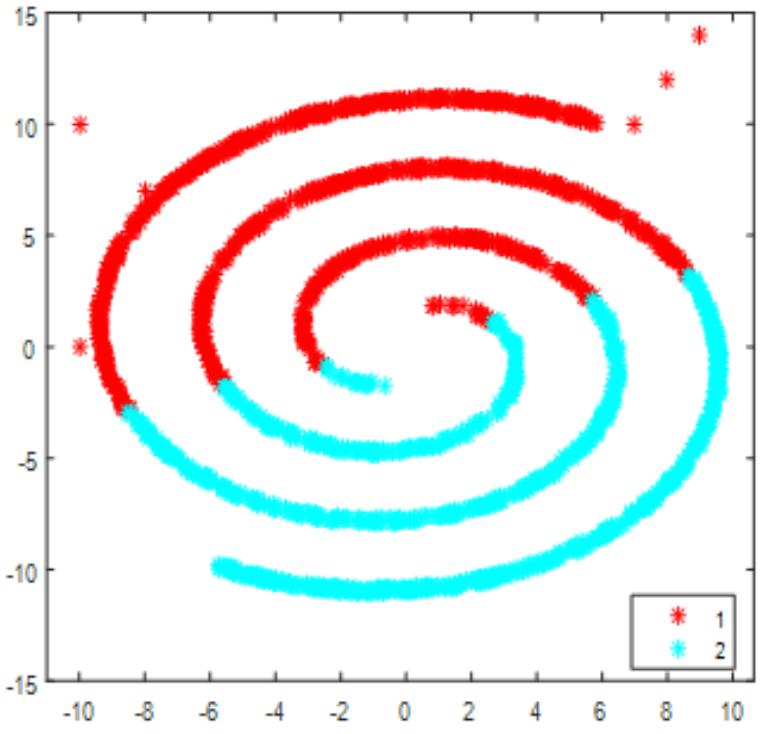


Eps = 10, MinPts = 4

K-Means vs DBSCAN



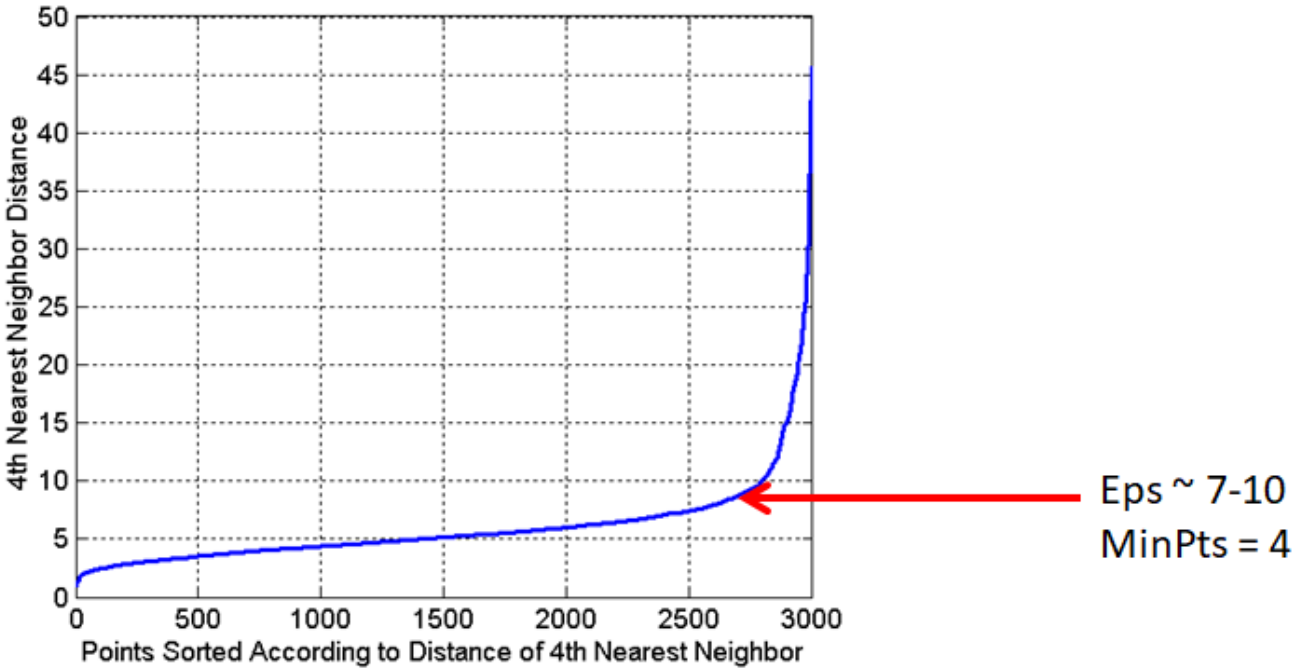
(a) DBSCAN clustering



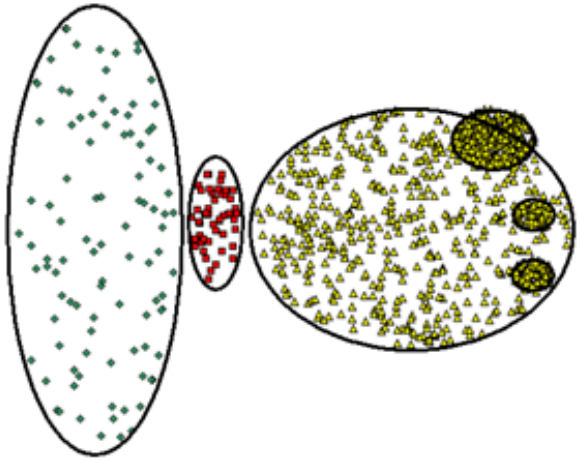
(b) K-Means Clustering

Determining Eps and MinPts in DBSCAN

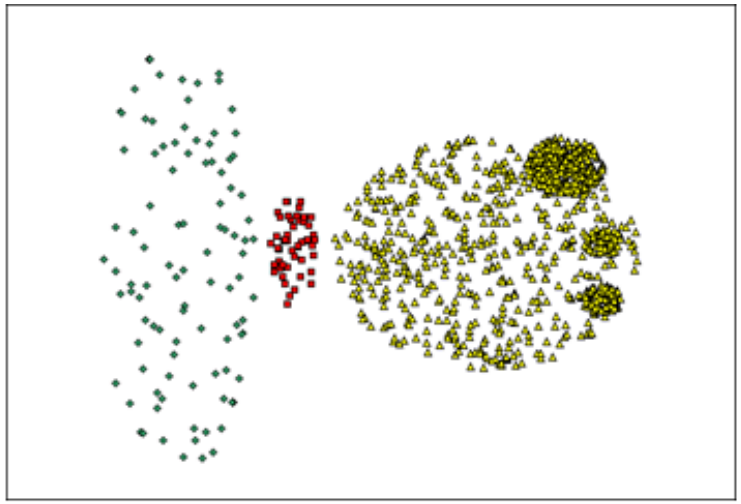
- Idea is that for points in a cluster, their k^th nearest neighbors are at roughly the same distance.
- Noise points have the kth nearest neighbor at farther distance.
- So, plot sorted distance of every point to its kth nearest neighbor.
- Find the distance d where there is a "knee" in the curve
  -Eps = d, MinPts = k



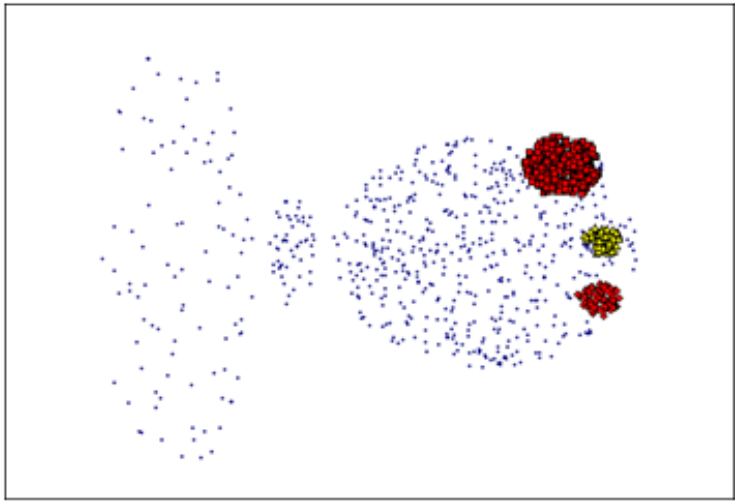
Drawback of Density based clustering



Original Points



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

- Varying densities
- High-dimensional data

DBSCAN: Sensitive to Parameters

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

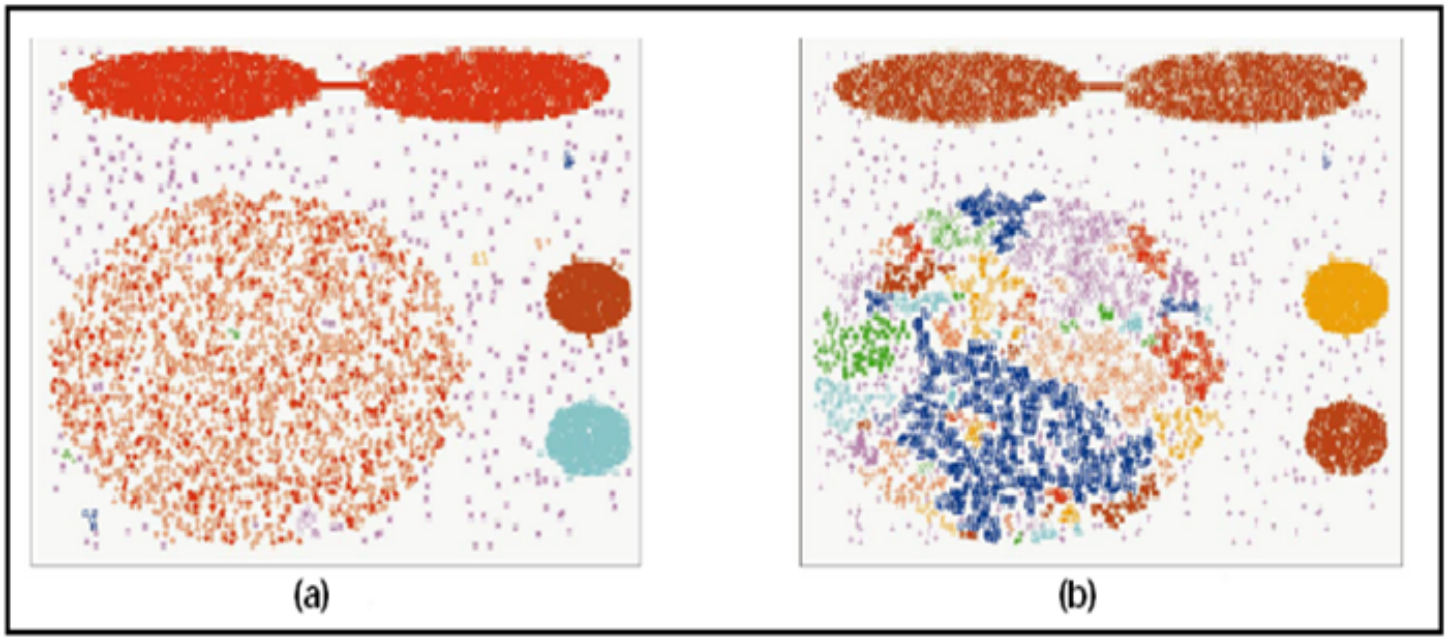
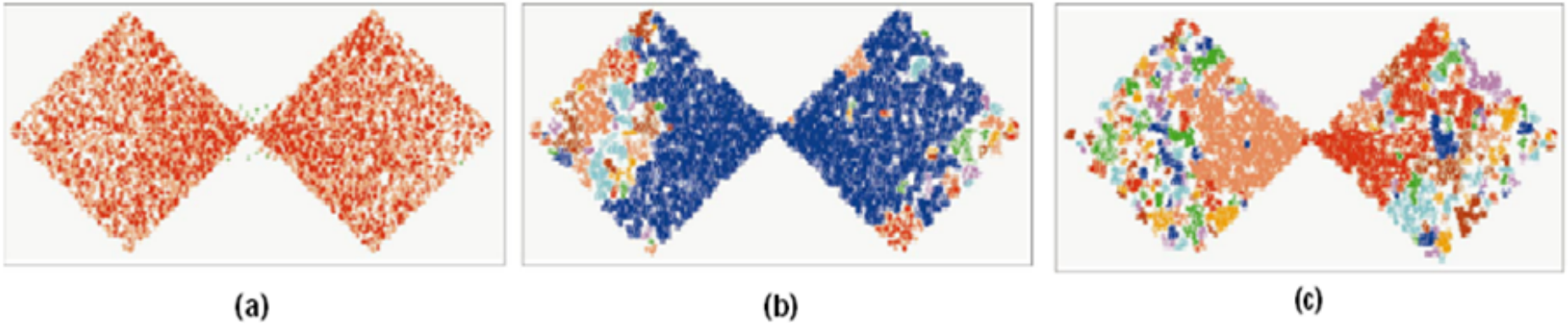



Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.